⚠ IMPORTANT — Support Scope

3DPTronics supports the VirtuaTilt controller hardware only. This guide is a *compendium* to help you connect your VirtuaTilt to third-party pinball software (VPX, Future Pinball + BAM, DOF/DOFLinx, PinEvent, PinUP Popper, etc.). We provide them as a courtesy to save you the hunt across multiple sites. **The information is best-effort** and kept reasonably current, but is **not** official documentation of those projects. **We do not provide official support for any third-party software referenced here** — for software-specific issues, refer to each project's own docs and community channels.

☑ Officially supported

Component	Coverage
🎮 **VirtuaTilt controller**	Wired and Wireless editions, all hardware components
💾 **VirtuaTilt firmware**	Updates and recovery via GUIConfigTool
📋 **Bundled profiles**	All Wired profiles shipped in the redist
⚙ **GUIConfigTool**	Configuration, calibration, profile loading, output testing
🔧 **Wireless Kit**	Wireless hardware kit for existing VirtuaTilt owners
📦 **Accessories**	Anything shipped directly by 3DPTronics

❌ Not supported

Category	Examples	Where to get help
🎲 **VP Game engines**	VPX, Visual Pinball, Future Pinball, BAM	VPForums · VP Universe
🕹 **VP titles**	Pinball FX / FX2 / FX Classic / VR / M, etc.	Zen Studios support
💥 **Feedback frameworks**	DOF (Direct Output Framework), DOFLinx	DOF on GitHub · DOFLinx wiki
🎨 **DMD / extras**	DMDExt, PinEvent V2, PuP-Packs / PUP-Stream	VP Universe
🚀 **Frontends**	PinUP Popper, Baller Installer, etc.	Popper forum
🎬 **Emulators / ROMs**	MAME, pinball ROMs	MAMEdev
💻 **System-level**	Windows, GPU/audio drivers, USB controllers	Microsoft / vendor support

🌐 Community help for software

Project	Community
Visual Pinball X	vpforums.org · vpuniverse.com
Future Pinball + BAM	VPU Future Pinball forum
Future Pinball + BAM AIO	VPU file 14807 — FP+BAM Essentials AIO
DOF	github.com/DirectOutput/DirectOutput
DOFLinx	doflinx.github.io/docs
Pinball FX family	Zen Studios Support
PinUP Popper	VPU PinUP Popper forum
MAME	mamedev.org

💡 **Why this matters:** Our team is small and our expertise is the VirtuaTilt hardware — we can't debug VPX scripts, flaky BAM installs, or broken DOF chains. Those communities have their own experts. The **AI chatbot on the wiki** can help connect VirtuaTilt with most of the software above; for deeper or vendor-specific issues, reach out to the appropriate community first.

VIRTUATILT – PinballFX VR Setup

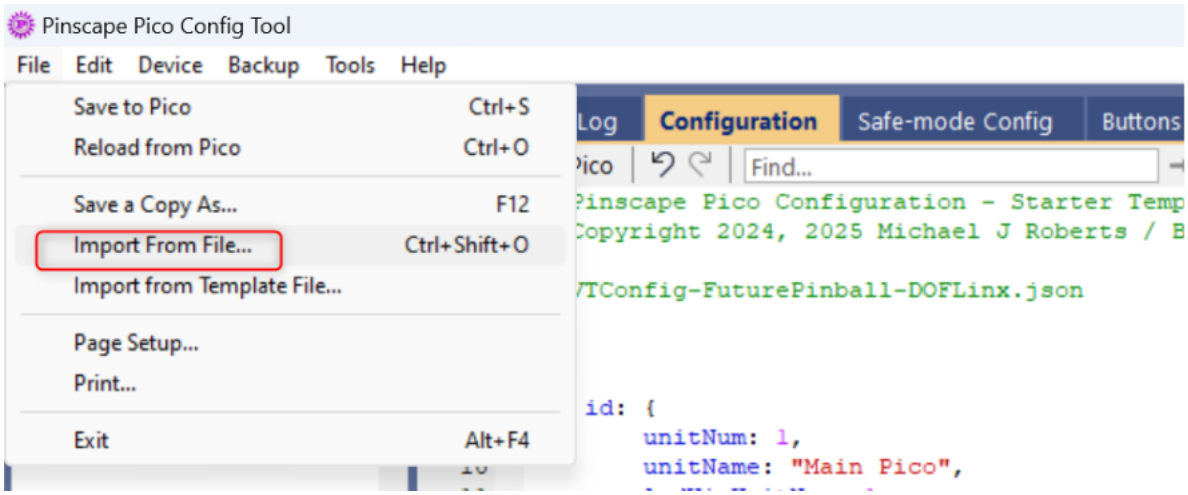
Following some basic instructions to have your new VirtuaTilt up and running with PinballFX VR.

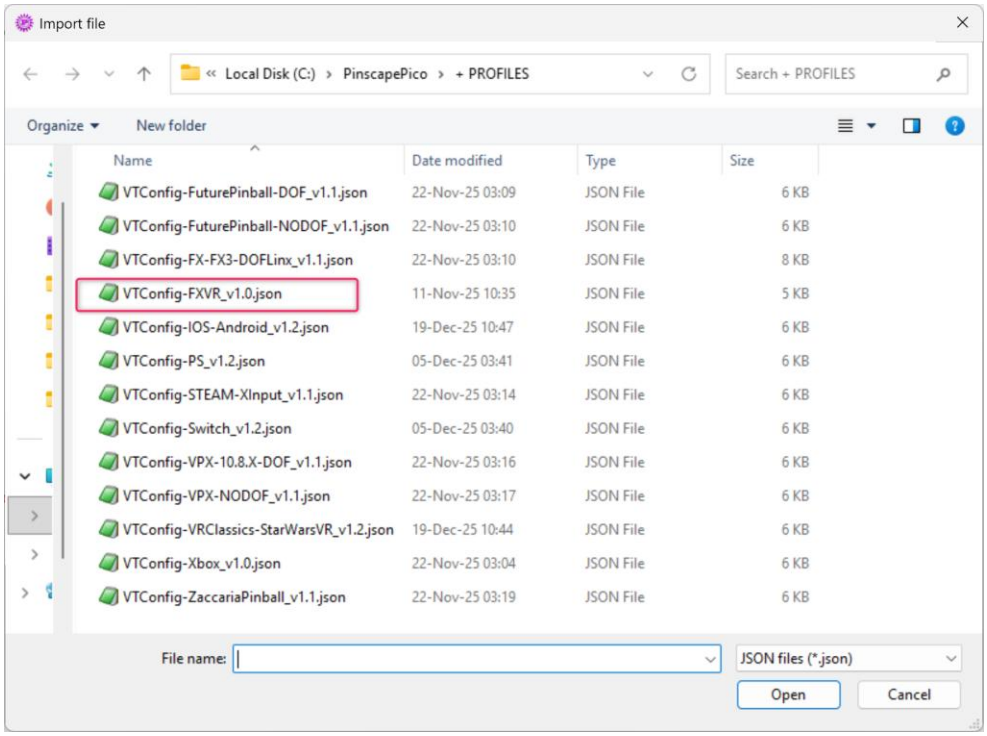
NOTE: You need an OTG adapter to directly connect your VirtuaTilt to Meta Quest (USB-A to USB-C)

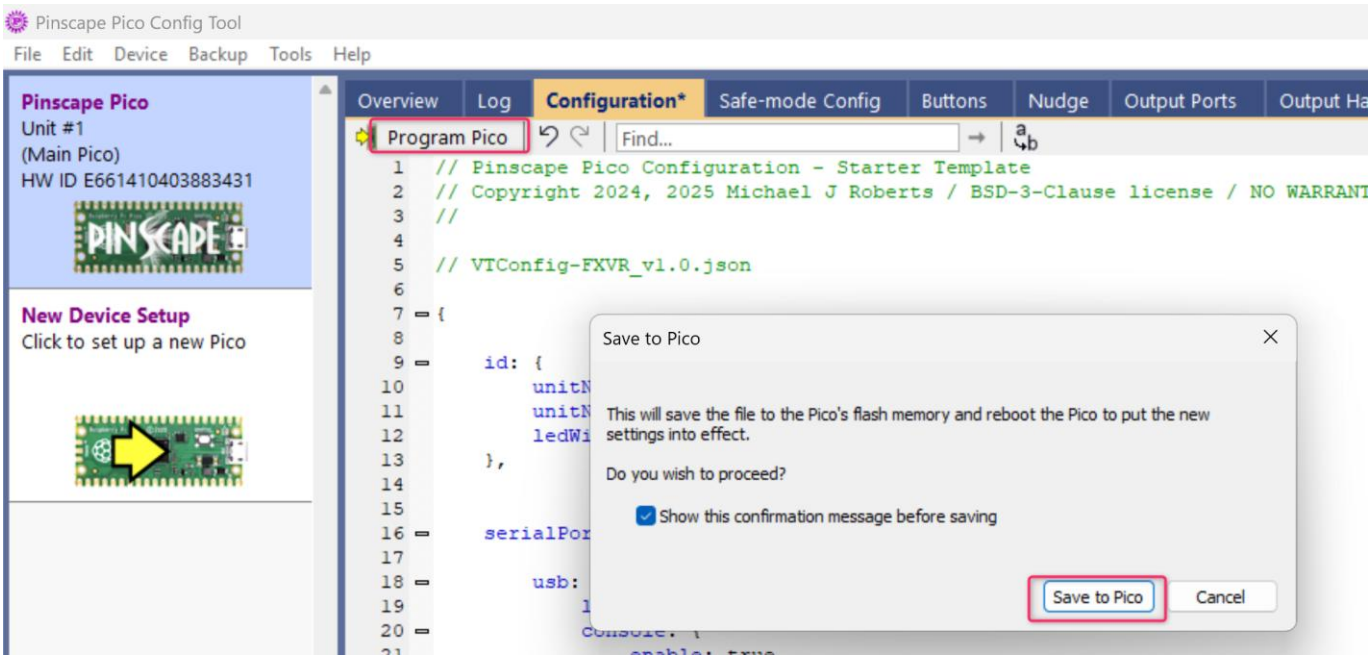
PREREQUISITES:

- **VTConfig-FXVR_XAM_v1.X.json** profile (v1.14 or later) loaded in **Pinscape Pico Config Tool**
- **VirtuaTilt firmware v8.0.20** (or later) installed (see **MAIN SETUP GUIDE – APPENDIX A**)

IMPORTANT NOTE: PinballFX VR 1.10 update (Judge Dredd) introduced two-stage flippers. VTConfig-FXVR_XAM_v1.14 profile moves both lower and upper flippers with each flipper button. Standard Mode profile (VTConfig-FXVR) is not provided anymore.

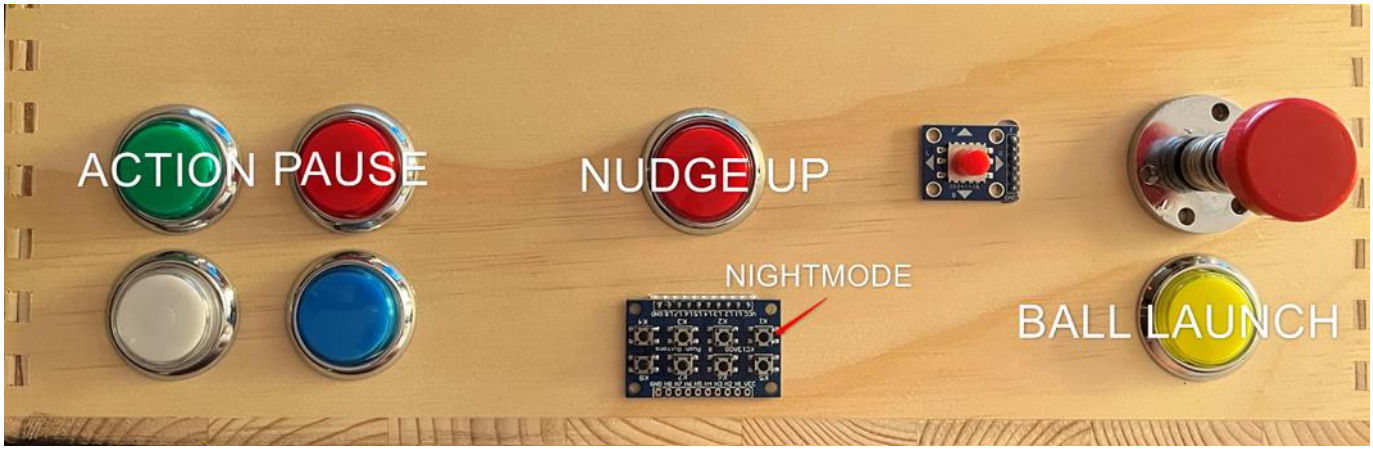


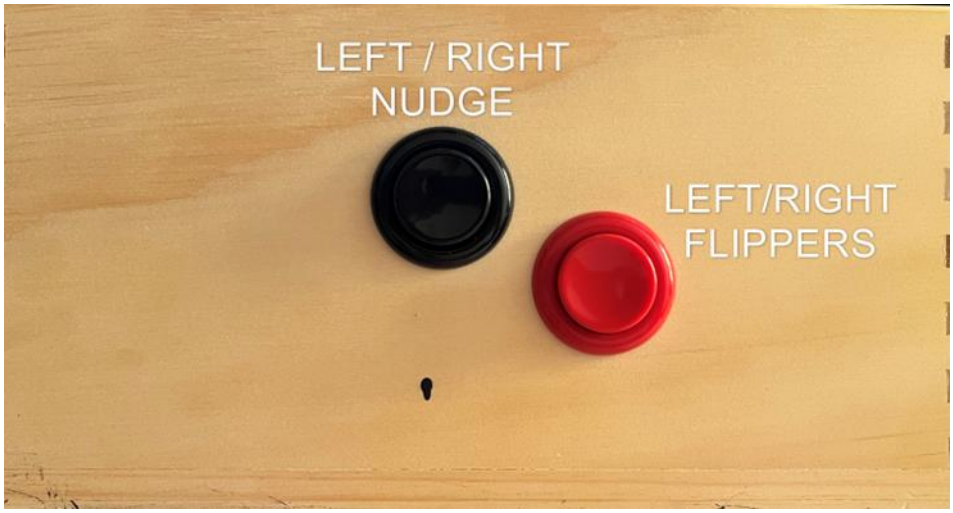

Import file
Local Disk (C:) › PinscapePico › + PROFILES
Search + PROFILES
Organize
New folder
Name
Date modified
Type
Size
VTConfig-FuturePinball-DOF_v1.1.json 22-Nov-25 03:09 JSON File 6 KB
VTConfig-FuturePinball-NODOF_v1.1.json 22-Nov-25 03:10 JSON File 6 KB
VTConfig-FX-FX3-DOFLinx_v1.1.json 22-Nov-25 03:10 JSON File 8 KB
VTConfig-FXVR_v1.0.json 11-Nov-25 10:35 JSON File 5 KB
VTConfig-IOS-Android_v1.2.json 19-Dec-25 10:47 JSON File 6 KB
VTConfig-PS_v1.2.json 05-Dec-25 03:41 JSON File 6 KB
VTConfig-STEAM-XInput_v1.1.json 22-Nov-25 03:14 JSON File 6 KB
VTConfig-Switch_v1.2.json 05-Dec-25 03:40 JSON File 6 KB
VTConfig-VPX-10.8.X-DOF_v1.1.json 22-Nov-25 03:16 JSON File 6 KB
VTConfig-VPX-NODOF_v1.1.json 22-Nov-25 03:17 JSON File 6 KB
VTConfig-VRClassics-StarWarsVR_v1.2.json 19-Dec-25 10:44 JSON File 6 KB
VTConfig-Xbox_v1.0.json 22-Nov-25 03:04 JSON File 6 KB
VTConfig-ZaccariaPinball_v1.1.json 22-Nov-25 03:19 JSON File 6 KB
File name:
JSON files (*.json)
Open
Cancel


Pinscape Pico Config Tool
File Edit Device Backup Tools Help
Pinscape Pico
Unit #1
(Main Pico)
HW ID E661410403883431
New Device Setup
Click to set up a new Pico
Overview Log Configuration* Safe-mode Config Buttons Nudge Output Ports
Program Pico
Find...
// Pinscape Pico Configuration - Starter Template
// Copyright 2024, 2025 Michael J Roberts / BSD-3-Clause license / NO WARRANT
// VTConfig-FXVR_v1.0.json
Save to Pico
This will save the file to the Pico's flash memory and reboot the Pico to put the new settings into effect.
Do you wish to proceed?
Show this confirmation message before saving
Save to Pico
Cancel

Buttons Mapping

This is default button mapping for PinballFX VR:





NOTE: **ACTION button is not used anymore** with VTConfig-FXVR_XAM_v1.14 profile (PinballFX VR 1.10 update).

Nudge Adjustments

Nudge is already configured and ready to use. You can nudge with left/right buttons **OR / AND** internal accelerometer.

IMPORTANT NOTE: **PinballFX VR does not support analog nudge natively.** However, VirtuaTilt is able to translate digital inputs into accelerometer movements!

You can adjust parameters for accelerometer-based nudging in **Pinscape Pico Config Tool** under **Configuration** tab:

```
// Nudge Right
{
    [redacted] onTime: "200", resetTime: "300", threshold: "22000"  },
},

// Nudge Left
{
    [redacted] onTime: "200", resetTime: "300", threshold: "22000"  },
},

// Nudge Up
{
    [redacted] onTime: "200", resetTime: "300", threshold: "18000"  },
```

onTime

The amount of time to pulse the imaginary button, in milliseconds (1000 milliseconds equals one second). When the nudge axis reading goes over the threshold, the button reads as pressed for this interval.

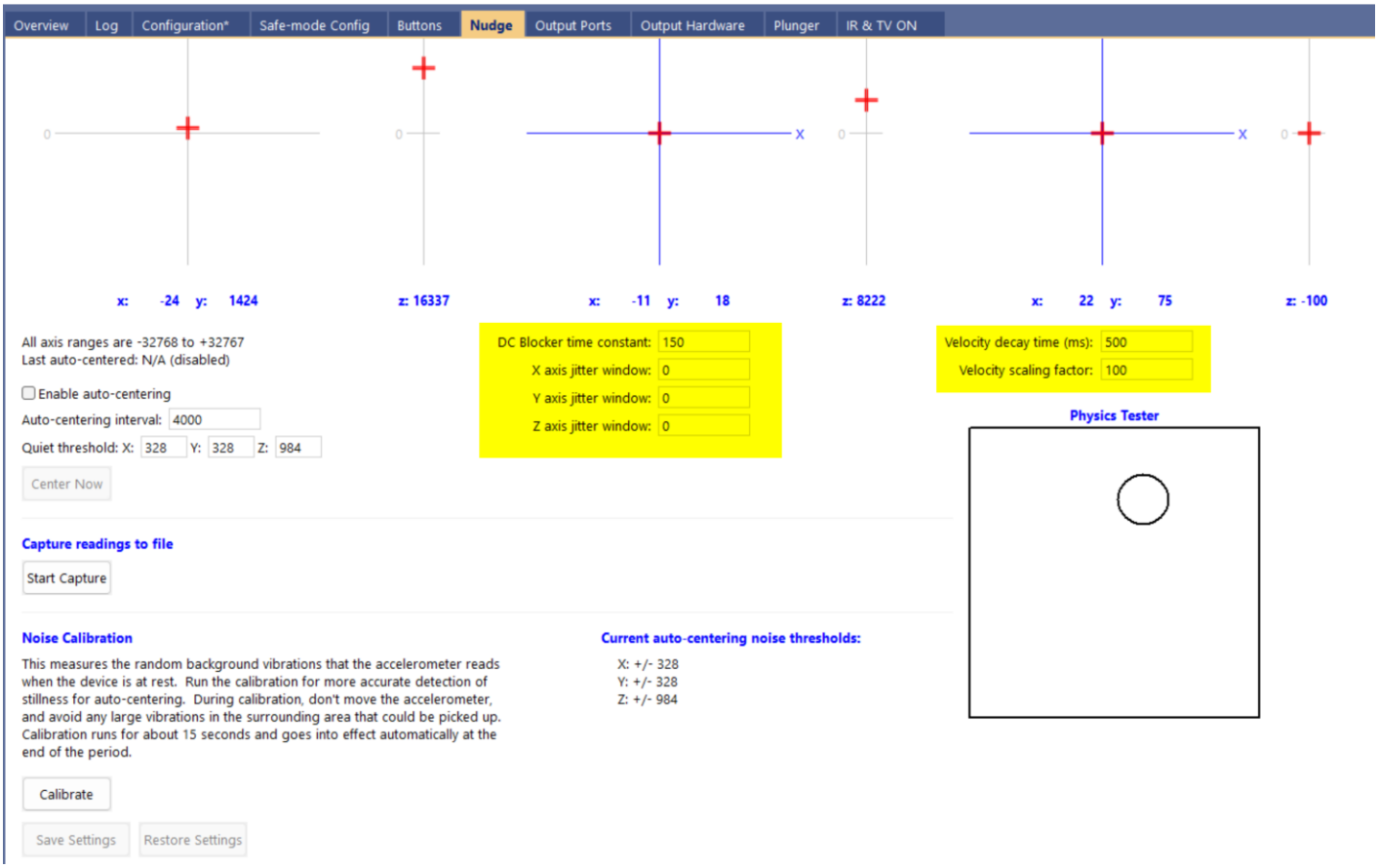
ResetTime

The reset time for the nudge event, in milliseconds (1000 milliseconds equals one second). Each time the button is triggered, it performs its action and then ignores the nudge input for this amount of time. This prevents the button from getting triggered repeatedly during a strong jolt to the cabinet.

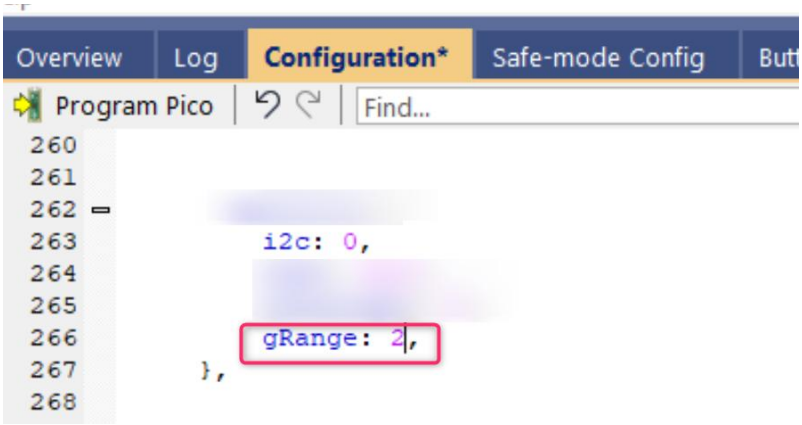
Threshold

The minimum reading on the nudge axis to trigger the event, 0 to 32767.

VERY IMPORTANT NOTE: You need to find your right settings (according to your personal preferences) for nudging. You can adjust both Pinscape Pico accelerometer settings and your VP game settings until you find your perfect spot:



You can also increase accelerometer sensitivity by changing gRange value to 2 in any profile:



Please refer to the official Pinscape Pico reference for additional information about accelerometer parameters: http://mjrnet.org/pinscape/PinscapePico/Help/NudgeWin.htm

Plunger Adjustments

IMPORTANT NOTE: **PinballFX VR does not support analog plunger natively; only ball launch button.** VirtuaTilt is able to translate a digital key press (ball launch button) into plunger movements, however it will never behave like a real analog plunger as in Pinball VR Classics or StarWars Pinball VR for example.

Plunger is already configured and ready to use. You can launch ball with button **OR / AND** mechanical plunger. You can adjust parameters for mechanical plunger in **Pinscape Pico Config Tool** under **Configuration** tab:



Threshold

Sets the analog reading at which the button will trigger, using the "normalized" range of 0 to 65535. The Pinscape software internally maps every device's native range to the same 0..65535 scale, where 0 corresponds the minimum voltage that the device can register, and 65535 corresponds to the maximum voltage.

X-Arcade Mode

To activate X-Arcade Mode in PinballFX VR you need **VirtuaTilt firmware v8.0.20** (or later) installed on your Pico, as described in **MAIN SETUP GUIDE – APPENDIX A**. **Custom firmware (PinscapePico_v1.X.X-XAM.uf2) is not needed anymore.**

You also need to import a new profile in Pinscape Pico config tool: **VTConfig-FXVR_XAM_v1.X.json** (v1.14 or later)

IMPORTANT NOTE: After a firmware update, import VTConfig-FXVR_XAM_v1.14 (or later) again, even if you were already using an older FXVR_XAM profile. Otherwise XAM won't be available in the game.

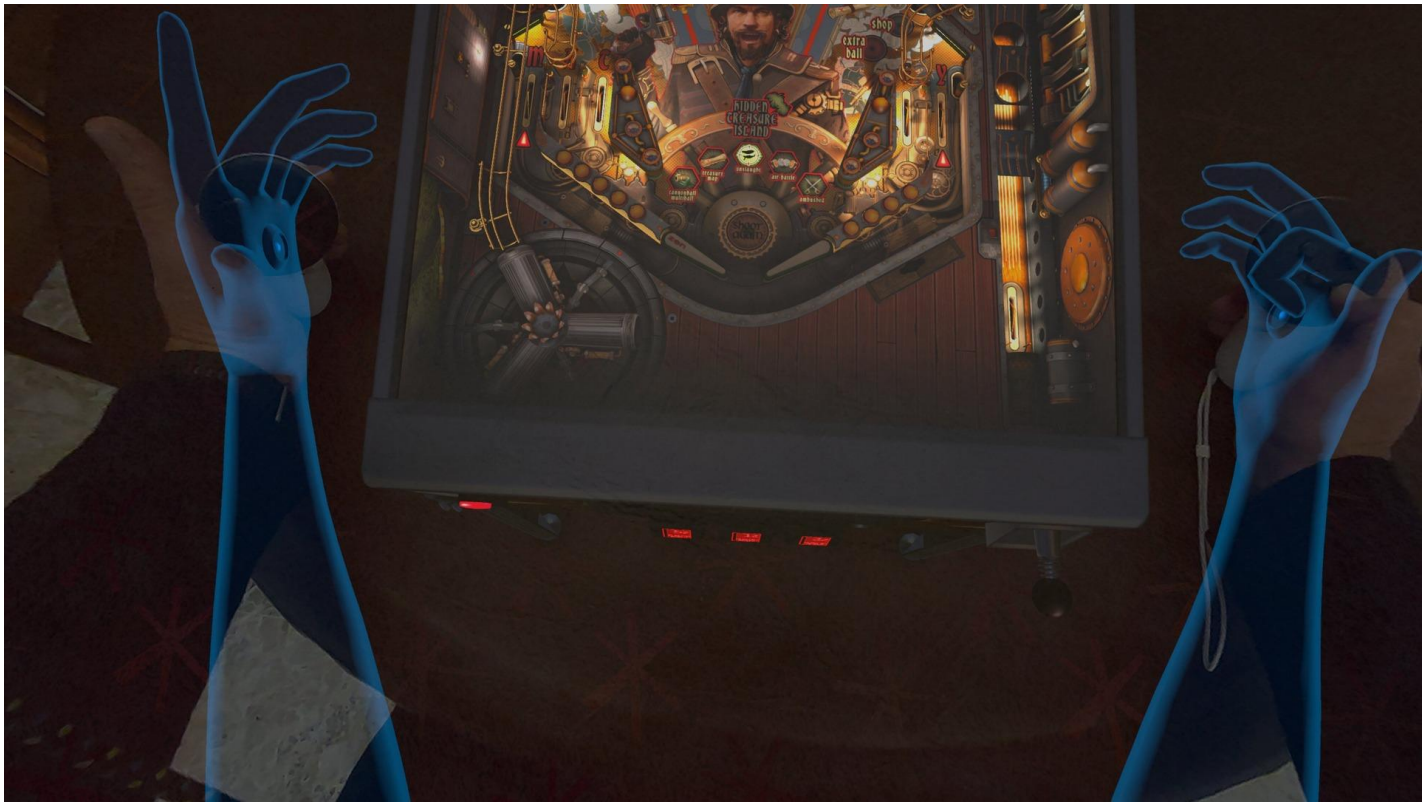
When you load the game, X-Arcade mode is automatically activated:



Now you can benefit from new in-game features:

- Use left/right flipper buttons to navigate menus (**Play again /Exit table**)
- Recenter table view by using Meta button on right Quest controller
- **Use digital plunger steps to behave like an analog plunger!**

To activate this game mode, you need to grab your Meta Quest controllers, align them vertically to left and right flipper buttons and then press left and right flipper buttons simultaneously:



IMPORTANT NOTE: **PinballFX VR 1.10 has a bug in XAM: after "Play again" left flipper doesn't work anymore.** Until Zen fixes it, select "Restart Table" from game menu.

Experience the full potential of VirtuaTilt with **PinballFX VR**!